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Advancement With AI in the Global Economy

Technology is growing at an unprecedented pace, and everywhere in the world is keeping an eye on it to foresee the impact on the global economy. Different from the previous technological advancement that focuses primarily on automating routine manual labor, the emergence of artificial intelligence represents a historical, structural shock to the overall global economic landscape. Today, the technology sector is no longer just a standalone vertical industry; it has become the horizontal foundation that supports global trade, capital flows, and complex decision-making in each business. In recent empirical studies conducted by researchers, many highlight that this recently emerged "AI shock" will be something that drastically alter both the labor markets and firm level productivity， having the potential to bringing in trillions of dollar in the economy while also severely threatening workforce displacement and widening inequality (Seamans & Raj, 2018) As each nation brace for the AI disruption, the global economic map is being redrawn anew, leading to a critical conjecture of how the world's two largest economies – the United States and China – choose to navigate its way through the age of AI, setting the stage for a new era of global competition.

Each country has its own approaches to the new AI era. The United States haschosen to embrace a decentralized, free-market model to harness the AI boom, allowing private-sector tech giants to drive rapid innovation in software to maximize capital efficiency and allowing the market force to decide the integration of AI models in business. In contrast, China is currently grappling with a "two-speed economy"—characterized by a slowing domestic property market alongside a surging high-tech manufacturing sector. To stabilize its economy and counter escalating Western trade barriers of tariffs and other obstacles, Beijing has deployed its 15th Five-Year Plan to regain control. The 15th Five-Year Plan seeks to bridge this economic divide by allocating resources to "New Quality Productive Forces" through prioritizing industrial modernization, achieving breakthroughs in bottleneck technologies like semiconductors, and securing new markets through global trade diversification. Through these strategic priorities, China aims to secure its geopolitical supply chains and fortify its economy against the vulnerabilities of traditional Western trade routes.

Given China's goal and its newly established 15th Five-Year Plan, a concern remains whether China's current two-speed economy can successfully manage a technological shock as unpredictable and dynamic as Artificial Intelligence. Many critics are arguing that the rigid, top-down economic strategies proposed by Beijing will stifle innovation and be unable to survive a technical shift of this magnitude. Despite the critics' criticism, China's 15th Five-Year Plan actually provides a superior, more resilient framework for managing the AI shock than the U.S. free-market model. Directly controlling capital toward a focus on critical hardware allows China's policy to correct industry resource misallocation, thus prioritizing long-term dynamic efficiency and geopolitical security over short-term corporate gains. Evaluating the capability and necessity of this strategy begins with recognizing the severe macroeconomic threats of the AI shock alongside the vulnerabilities of the U.S. response. To review this case of effectiveness, recent empirical evidence on industrial policy (Li et al., 2026) demonstrates that state-directed capital allocation can successfully foster long-term productivity enhancements. Ultimately, China’s strategy of anchoring its strategy towards new technical innovations would effectively absorb the AI shock, ensuring domestic stability and a durable path to global technological leadership.

To understand the necessity of state intervention in the modern era, one must first define the parameters of the AI shock and how it's different from past technical advancements. The current wave of Artificial Intelligence is replacing brainpower and decision-making. Tech isn't just a single standalone industry anymore; it has become a horizontal foundation that supports many fields like healthcare and finance, thus supporting all economic activity. Today, data and insights have become an important commodity for any business. The AI shock thus fundamentally restructures how capital and labor interact, altering the baseline requirement for any businesses in the market.

The macroeconomic incentives driving the rapid adoption of Artificial Intelligence are unprecedented for any country. According to projections by McKinsey, the deployment of Generative AI could inject between $2.6 trillion and $4.4 trillion into the global economy annually—an astronomical amount roughly equivalent to the entire Gross Domestic Product of the United Kingdom. Furthermore, the Digital Cooperation Organization forecasts that the market value of AI-driven software development will reach $207 billion by 2030. Due to the AI’s capability of turning vast amounts of raw data into immediate, actionable insights, it acts as an unprecedented multiplier for capital efficiency for any economy. A conclusion was thus reached that, for the corporate world, the financial upside of Artificial Intelligence is just too massive to ignore.

Technological advancement always brings new risks, and the rapid growth of the tech sector is no exception. Despite the technology sector’s growth, offering any business a path to increased productivity, it also threatens to widen the "digital divide" and worsen wealth inequality. High-tech regions often concentrate economic growth, leaving areas with less digital infrastructure struggling to keep pace. Empirical research reveals that it fundamentally destabilizes labor markets. Seamans and Raj (2018) demonstrate that unregulated automation actively displaces workers and depresses wages across vulnerable sectors. This is the fatal flaw of relying on an unregulated free market to manage the AI shock. Without strict government intervention to distribute these gains, the rapid integration of the newly emerged intelligent systems will carve a massive "digital divide" through the economy. Leading to the wealth generated by AI being pooled exclusively toward the top high-tech professionals, leaving the average worker behind as the rest of the workforce gets displaced by software.

In response to the heavy economic change that the AI shock brings along, the United States is relying on a decentralized, free-market framework. Rather than implementing top-down industrial policy, the U.S. model delegates capital allocation to the private sector, thus allowing Silicon Valley to dictate the pace and direction of AI integration. By focusing almost exclusively on assets like generative software and cloud computing, the framework is designed to maximize speed and corporate efficiency above all else. Although this unregulated pace allows for explosive growth and fast innovation in the short term, it does not look at the long-term structural economic stability, as the Government lacks control over where the economy is actually heading.

The immediate result of this free market approach is a profound concentration of corporate wealth among the wealthy. On the surface, the U.S. free-market strategy appears wildly successful. By 2024, the dominance of the technology sector had become so heavily skewed that just eight major tech firms—most notably AI infrastructure leaders like Nvidia and

Microsoft— were responsible for a staggering 55% of all U.S. stock market gains (Albrecht, 2025). This result is significant and is not representative of a healthy and diversified economy; instead, it showcases the fragile state of a system where the financial stability of the entire nation is dangerously dependent on the speculation of success from a few AI monopolies. An economy that arrives in this state is often highly susceptible to market volatility and speculative bubbles, such that the economy could easily crash if the AI bubble pops.

The most critical failure of the U.S. free market model lies in its inability to manage the socioeconomic fallout of rapid automation coming from the AI shock. This change can already be seen in the severe labor market restructuring, as brutal tech layoffs are happening everywhere. As each company is entirely focused on winning the AI race, they are treating their own employees like unnecessary expenses. This can be seen from tech giants like Meta continuously executing waves of layoffs, gutting entire non-AI departments simply to redirect billions of dollars toward AI infrastructure. Despite the actions conducted by major AI firms to win the AI race, the financial foundation of this AI boom remains highly fragile. This can be seen in the case of OpenAI, the pioneer of the generative AI market, which is reportedly projected to face up to $5 billion in annual operating losses, driven by the staggering multibillion-dollar computational costs required to train large language models. This situation reveals that by leaving the AI transition entirely to the free market, the U.S. has created a fragile bubble that actively destroys tens of thousands of jobs while burning through a high amount of capital.

The stark contrast of the U.S. model, where the Government delegates capital allocation to the private sector as they drive innovation, while massive layoffs are happening and being written off as the cost of innovation, is a system that is ill-managed to traverse the AI shock. This reality leads to a model where surviving the AI shock requires deliberate state intervention rather than unregulated corporate speculation and leaving it to corporate decision-making. This is the rationale driving China's current economic strategy, where it designed and employs a controlled, state-directed plan to absorb the AI shock. China is deploying its 15th Five-Year Plan to construct a rigid and flexible plan that focuses on prioritizing long-term structural resilience and a stable economy over short term market spikes.

To understand the strategic architecture of the 15th Five-Year Plan, examining the macroeconomic imbalances currently defining the current state of China is a must. China is currently operating as a “two-speed economy.” On one side, domestic consumption remains suppressed due to a persistent property sector crisis. This is because the majority of Chinese household wealth is centered in real estate, and falling property values have triggered a negative wealth effect, causing families to cut back on spending frequency (PricewaterhouseCoopers, 2024) The fallen property value is not just small, it is to the extent of slowing GDP growth by as much as two percentage points as residential investment fell 15.9% year-on-year. On the other hand, China’s industrial sector is thriving, evidenced by a record trade surplus of nearly $1.2 trillion in 2025, and a 6.6% year-on-year rise in December exports driven by high-tech manufacturing. (Goldman Sachs, 2025) Recognizing this unsustainable imbalance, Beijing plans on utilizing the 15th Five-Year Plan to aggressively intervene as a foundational blueprint for a strategic roadmap that will bridge this divide by prioritizing industrial modernization and technological self-sufficiency to solidify a more resilient economic model by transitioning away from the fragile real estate market to one that actively works on advanced technical infrastructure to achieve stability.

The central mechanism of the 15th Five-Year Plan is centered on the "New Quality Productive Forces." Rather than focusing purely on software innovation and intangible assets like the United States, Beijing's strategy is to priortize high quality industrial modernization that makes AI possible, while not leaving the economic decision purely to the private sector. The state is planning to channel capital to overcome vulnerabilities in "bottleneck technologies" like advanced semiconductors, lithium-ion batteries, and massive green energy grids (Institute for China-America Studies, 2025). The logic stemming from these decisions comes from the computational demands the AI shock requires for its massive physical infrastructure, as the AI revolution is useless without the physical infrastructure to power it. Thus, China is laying out a foundation by actively securing the base layer of the technological supply chain, such that the AI integration is supported by industrial capacity for the future.

China's strategy from its 15th Five-Year Plan extends beyond its domestic border by actively using its technological advancement in its geopolitical region. Beijing, recognizing the increasing Western trade barriers, escalating U.S. tariffs, and U.S. technological containment, is focusing on diversifying its supply chain. By leveraging its dominance in green technologies and hardware, China is establishing cost-effective, secure logistics networks across developed nations in Asia, Latin America, and Africa (Boata et al., 2025). This growing regional concentration of trade strengthens China's supply chain resilience, allowing China to scale new networks while successfully bypassing Western sanctions. This ensures that even if the U.S. attempts to sever China from the global tech economy, Beijing has already secured a massive market in the Global South to sustain its AI and manufacturing sectors.

Critics of China's state-led model frequently point to severe industrial overcapacity and "involution"—excessive competition that drives down corporate profitability—as evidence of policy failure. Then, following up by arguing that the U.S. free market is superior because it avoids state-sponsored overcapacity is a misunderstanding of how this new technology is changing the economy. Chinese economic planners indeed acknowledge the issue of "involution". Yet, empirical data prove that this short-term notion is a necessary sacrifice for long-term dominance. Econometric analysis of China's industrial policy reveals that state intervention successfully redirects massive capital inflows to previously undercapitalized, high-value industries. While this does create static inefficiencies in the short term, this motion can lead to massive "dynamic efficiency gains and long-term productivity enhancements" (Li et al., 2026). By utilizing the 15th Five-Year Plan to tackle this involution, Beijing is "promoting 'consolidation,' meaning fewer producers, higher value-added output, and stronger national champions" (Wang, 2026). Ultimately, by doing so, China is effectively building a strong dominance in AI technologies by sacrificing short-term corporate profits.

The U.S. model contains a critical vulnerability in its reliance on Wall Street to dictate the pace of the AI revolution. Meanwhile, China’s 15th Five-Year Plan guarantees stability by intentionally "revitalizing finance without liberalizing it" (Zhang, 2025). The U.S. is allowing the AI-market to inflate stock markets without building the underlying industrial capacity, which Beijing has objected to doing. Beijing considers U.S. decisions as a wrongful approach because "unfettered financialization is seen as a strategic risk" (Zhang, 2025). The Chinese framework operates on a critical realization: finance must serve actual industry, which is why the state is "curbing speculative, high-risk finance and simultaneously channeling capital and infrastructure behind strategic, state-led financial innovation" (Zhang, 2025). Thus, by tightly regulating capital and shifting investment toward self-sufficiency, China aims at "seizing the commanding heights of technological development "(Zhu, 2025), ensuring that its technological leap is grounded in real economic value rather than a fragile speculation from the financial market.

Ultimately, China's 15th Five-Year plan presents a sufficiently strong model that aligns itself with geopolitical reality, ensuring that it lays a strong foundation for the future. The plans fundamentally evolved their strategy; technological self-reliance is no longer a reactive defense mechanism against U.S. tariffs, but a proactive driver of its entire economy. By redirecting capital away from the property sector and into frontier technologies like AI, the Chinese leadership has chosen a path of high-quality development. This roadmap would allow Beijing to absorb and sustain in the AI era, avoiding the fatal socioeconomic flaws of the American free market, and solidify a path for its future success.

Works Cited:

Allianz. Trade Routes in an Era of Geopolitics: Navigating the New Global Map. 5 Nov. 2025, www.allianz.com/content/dam/onemarketing/azcom/Allianz_com/economic-research/publications/specials/en/2025/november/2025_11_05_Trade-Routes-AZ.pdf.

Asia Pacific Foundation of Canada. China's Economic Strategy for 2026: Prioritizing Continuity over Change. 2025, www.asiapacific.ca/publication/chinas-economic-strategy-2026-prioritize-continuity-over-changE.

Boata, Ana, et al. Trade Routes in an Era of Geopolitics. Allianz Research, 2025.

"China's Economy in November 2025: Year-end Review and 2026 Outlook." China Briefing, 2025, www.china-briefing.com/news/chinas-economy-in-november-2025-year-end-review-and-2026-o utlook/.

"China's Fifteenth Five-Year Plan: Stability, Modernization, and the Strategic Logic Behind its Domestic Priorities." Institute for China-America Studies, 2025, chinaus-icas.org/research/chinas-fifteenth-five-year-plan-stability-modernization-and-the-strategic-logic-behind-its-domestic-priorities/.

"De-risking and Reconnecting: Europe, China, and Africa's Green Transition." ECDPM, 2025, ecdpm.org/events/de-risking-reconnecting-europe-china-africa-green-transition.

Goldman Sachs. China's Economy Expected to Grow in 2026 Amid Surging Exports. 2025, www.goldmansachs.com/insights/articles/chinas-economy-expected-to-grow-in-2026-amid-surgiNg-exports.

JP Morgan Asset Management. Path to a High-Quality and More Balanced Growth: What to Expect from China's Fifteenth Five-Year Plan. 2025, https://am.jpmorgan.com/tw/en/asset-management/institutional/insights/market-insights/market-updates/on-the-minds-of-investors/path-to-a-high-quality-and-more-balanced-growth-what-to-expect-from-chinas-fifteenth-five-year-plan/

PricewaterhouseCoopers. China Economic Quarterly Q4 2024. 2024, www.pwccn.com/en/research-and-insights/china-economic-quarterly-q4-2024.pdf.

Digital Cooperation Organization. Digital Economy Trends 2024. Digital Cooperation Organization, 2024, https://dco.org/wp-content/uploads/2024/10/Digital-Economy-Trends-2024.pdf.

Huawei. "Global Digitalization Index (GDI) 2024." Huawei, 2024, https://www.huawei.com/en/gdi.

"The Economic Potential of Generative AI: The Next Productivity Frontier." McKinsey & Company, 14 June 2023, https://www.mckinsey.com/capabilities/tech-and-ai/our-insights/the-economic-potential-of-generative-ai-the-next-productivity-frontier

Albrecht, Bella. "Tech and Communication Stocks Drove US Market Gains in 2024." Morningstar, 3 Jan. 2025, https://global.morningstar.com/en-gb/markets/tech-communication-stocks-drove-us-market-gains-2024

Seamans, Robert, and Manav Raj. "AI, Labor, Productivity and the Need for Firm-Level Data." *National Bureau of Economic Research*, Working Paper no. 24239, Jan. 2018, pp. 1-13. *NBER*, http://www.nber.org/papers/w24239.

Li, Hui, et al. "Industrial Policy and Capacity Utilization: Evidence from China's Five-Year Plans." *International Journal of Industrial Organization*, vol. 104, 2026, article 103237. *ScienceDirect*, https://ideas.repec.org/a/eee/indorg/v104y2026ics0167718725001031.html

Olatunbosun, Adeyemi, et al. "How Artificial Intelligence Is Reshaping Labor Productivity: A Narrative Review of Theoretical Perspectives and Emerging Empirical Evidence." Journal of Economics, Business, and Commerce (JEBC), vol. 2, no. 2, 2025, pp. 414-425. Stecab Publishing, https://doi.org/10.69739/jebc.v2i2.1333.